

Personal Finance Coaching Database

Final Project Report

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1. Introduction and Mini World

For Project 1 in CIS 344 I chose the the Personal Finance Coaching Miniworld. I designed and built a database for a personal finance coaching business. A firm that pairs each client with a financial coach, helps clients set savings goals, and tracks the money coming in and going out. I chose it because I could picture it clearly, and because it has a good mix of people (coaches and clients) and records (goals and transactions). Four tables is enough for real relationships and foreign keys, but small enough to build carefully.

2. How I Determined the Requirements

I did not interview anyone or run a survey. I brainstormed the requirements from my own experience as a gym-goer and from what I understand about how personal trainers work with their clients. A financial coach and a trainer do a similar job: they are the expert, they work one-on-one, and they help someone reach a goal and stay on track. So I matched the two worlds:

In the gym	In my coaching business
Personal trainer	Coach
Member who trains with a trainer	Client
Fitness goal with a target and progress	Savings goal with a target amount, current amount, and date
Workout log or check-ins	Transactions (income and expenses)

From there, my process was:

1. **Start with the two parties.** I begin with the coach and the client and asked what the firm would need to know about each. A trainer might specialize in strength or weight loss, so a coach gets a Specialty.
2. **Ask what they track together.** Two things came up right away: the client's goals and the client's money. That gave me savings goals and transactions.
3. **Choose the fields.** Goals get a target amount, current amount, and target date. Transactions get a category, amount, date, and a Type (Income or Expense).
4. **Set the relationships.** A trainer has many members, but a member has one main trainer, so one coach has many clients. A client has many goals and many transactions. Then I checked that every table had a primary key and every foreign key pointed at something real.

I kept the scope to these four tables on purpose. Coaching sessions, appointments, payments to the firm, and clients with more than one coach are left out.

3. System Design

Table	Fields (PK = primary key, FK = foreign key)
Coaches	Coach_id (PK), First_name, Last_name, Specialty, Email
Clients	Client_id (PK), First_name, Last_name, Email, Phone, Coach_id (FK)
SavingsGoals	Goal_id (PK), Client_id (FK), Goal_name, Target_amount, Current_amount, Target_date
Transactions	Transaction_id (PK), Client_id (FK), Category, Amount, Transaction_date, Type

All three relationships are one-to-many: one Coach guides many Clients, one Client sets many SavingsGoals, and one Client has many Transactions. The foreign keys are Clients.Coach_id, SavingsGoals.Client_id, and Transactions.Client_id.

4. ER Diagrams

Chen style (by hand). Entities are rectangles (Coach, Client, SavingsGoal, Transaction), attributes are ovals with the primary key underlined, and relationships are diamonds (Guides, Sets, Has). The 1 and N next to each entity show the one-to-many side. UML style (MySQL Workbench). Each table is a box with its columns listed, and the 1:n non-identifying relationship tool draws the crow's foot lines and adds the foreign key columns. Both diagrams show the same design: the Chen diagram is the plan, and the Workbench diagram is that plan built as real tables. The images are in the repository (docs/chen_er_diagram.jpg and docs/eer_diagram.png).

5. Building the Database

I created the four tables in an EER diagram in MySQL Workbench, connected them with 1:n relationships, renamed the automatic foreign key columns to Coach_id and Client_id, and used Forward Engineer to generate the SQL. IDs are INT, text is VARCHAR(45), Phone is VARCHAR(20), money is DECIMAL(10,2), and dates are DATETIME. The foreign keys use NO ACTION, so a coach with clients (or a client with goals or transactions) cannot be deleted by accident.

Two scripts are included. finance_coaching_create_table.sql builds the database and tables. personal_finance_coaching_data.sql loads sample data (3 coaches, 6 clients, 7 goals, and 12 transactions) and runs example queries, such as savings goals per client, income versus expenses, coaches by specialty, and a join that lists each coach with their clients.

6. Challenges and How I Solved Them

- **The diagrams were the hardest part.** This is where I had the least experience. I did not understand what each shape meant, where the 1 and N go, or how a diamond turns into a foreign key. I treated the Chen diagram as the plan and the Workbench diagram as the built version, and checked each against my requirements until they matched.
- **Consistent names.** My first sketch used different column names than my requirements document, so I redrew it and rebuilt the tables until every name matched.
- **Data types.** Workbench's defaults were wrong for some columns. Phone was INT, which cannot hold dashes, and the amounts were VARCHAR, which would not add up. I changed them to VARCHAR(20) and DECIMAL(10,2).
- **Foreign keys and running the script.** The relationship tool creates long automatic column names and needs the "many" table clicked first, so I renamed the columns afterward. At first I used the run button that only runs one line, so my tables did not appear. I also fixed a typo in the schema name and made my query table names match the capitalization of the tables so they work on other systems.

7. Outcomes and What I Learned

The database works as designed. After loading, the row counts matched the data I inserted, the queries returned the results I expected, and the coach-and-client join listed every coach with their two clients, which shows the foreign keys connect the tables correctly. The biggest lesson was that planning first saves trouble: when the requirements, diagrams, and tables all use the same names and structure, everything else goes smoothly. If I kept building, I would add a sessions table, and monthly budgets by category.

8. Repository

All project files are in a public GitHub repository: docs/ (requirements, this report, and both diagrams), sql/ (both scripts), model/ (the .mwb file), and a README.md that explains how to run everything.

Project 1 Chen style diagram

